

Supplementary Evidence for the Corrected CI Decision Simulation

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6 September 2026

Scope

This supplement replaces all numerical tables and figures from the original submission. The source-of-truth file and machine-readable artifacts define the corrected numerical record. Original materials are retained under review/original. These are simulation experiments, not causal release-cost estimates.

Full replay conditions

artificial default delay

Policy	Mean cost	Sample SD	95% seed CI
Always block	1,865	0	1,865 to 1,865
Always canary	2,020	0	2,020 to 2,020
Always deploy	2,900	0	2,900 to 2,900
Bayesian rate	1,713	0	1,713 to 1,713
Rolling cost rule	2,085	0	2,085 to 2,085
Heuristic	2,288	0	2,288 to 2,288
LinUCB	1,877.5	0	1,877.5 to 1,877.5
Bias-only LinUCB	1,735	0	1,735 to 1,735
LinUCB wrapper	1,877.5	0	1,877.5 to 1,877.5
Static rules	1,929	0	1,929 to 1,929
Thompson	1,889.6	52.9	1,871 to 1,908.7

online smoke

Policy	Mean cost	Sample SD	95% seed CI
Always block	1,865	0	1,865 to 1,865
Always canary	2,020	0	2,020 to 2,020
Always deploy	2,900	0	2,900 to 2,900
Bayesian rate	1,713.5	0	1,713.5 to 1,713.5
Rolling cost rule	1,970.5	0	1,970.5 to 1,970.5
Heuristic	2,325	0	2,325 to 2,325
LinUCB	1,863	0	1,863 to 1,863
Bias-only LinUCB	1,739	0	1,739 to 1,739
LinUCB wrapper	1,863	0	1,863 to 1,863
Static rules	1,884	0	1,884 to 1,884
Thompson	1,857.1	62.6	1,835.2 to 1,878.5

real github actions

Policy	Mean cost	Sample SD	95% seed CI
Always block	1,027	0	1,027 to 1,027
Always canary	836	0	836 to 836
Always deploy	860	0	860 to 860
Bayesian rate	669	0	669 to 669
Rolling cost rule	624.5	0	624.5 to 624.5
Heuristic	860	0	860 to 860
LinUCB	779	0	779 to 779
Bias-only LinUCB	650	0	650 to 650
LinUCB wrapper	779	0	779 to 779
Static rules	640	0	640 to 640
Thompson	665.7	39.8	652.1 to 680.1

robustness high failure

Policy	Mean cost	Sample SD	95% seed CI
Always block	1,865	0	1,865 to 1,865
Always canary	3,180	0	3,180 to 3,180
Always deploy	5,800	0	5,800 to 5,800
Bayesian rate	1,865	0	1,865 to 1,865
Rolling cost rule	2,425.5	0	2,425.5 to 2,425.5
Heuristic	4,115	0	4,115 to 4,115
LinUCB	1,972	0	1,972 to 1,972
Bias-only LinUCB	1,905	0	1,905 to 1,905
LinUCB wrapper	1,972	0	1,972 to 1,972
Static rules	2,578	0	2,578 to 2,578
Thompson	1,955.8	27.2	1,946.1 to 1,965.5

robustness long delay

Policy	Mean cost	Sample SD	95% seed CI
Always block	1,865	0	1,865 to 1,865
Always canary	2,020	0	2,020 to 2,020
Always deploy	2,900	0	2,900 to 2,900
Bayesian rate	1,713	0	1,713 to 1,713
Rolling cost rule	2,077	0	2,077 to 2,077
Heuristic	2,319	0	2,319 to 2,319
LinUCB	1,973	0	1,973 to 1,973
Bias-only LinUCB	1,777.5	0	1,777.5 to 1,777.5
LinUCB wrapper	1,973	0	1,973 to 1,973
Static rules	1,924	0	1,924 to 1,924
Thompson	1,917	61.6	1,895.5 to 1,939.3

robustness low block

Policy	Mean cost	Sample SD	95% seed CI
Always block	1,005	0	1,005 to 1,005
Always canary	2,020	0	2,020 to 2,020
Always deploy	2,900	0	2,900 to 2,900
Bayesian rate	1,005	0	1,005 to 1,005
Rolling cost rule	1,273.5	0	1,273.5 to 1,273.5
Heuristic	2,325	0	2,325 to 2,325
LinUCB	1,072	0	1,072 to 1,072
Bias-only LinUCB	1,024	0	1,024 to 1,024
LinUCB wrapper	1,072	0	1,072 to 1,072
Static rules	1,591	0	1,591 to 1,591
Thompson	1,069.6	20	1,062.8 to 1,076.9

robustness short delay

Policy	Mean cost	Sample SD	95% seed CI
Always block	1,865	0	1,865 to 1,865
Always canary	2,020	0	2,020 to 2,020
Always deploy	2,900	0	2,900 to 2,900
Bayesian rate	1,714.5	0	1,714.5 to 1,714.5
Rolling cost rule	1,949	0	1,949 to 1,949
Heuristic	2,294	0	2,294 to 2,294
LinUCB	1,950	0	1,950 to 1,950
Bias-only LinUCB	1,750	0	1,750 to 1,750
LinUCB wrapper	1,950	0	1,950 to 1,950
Static rules	1,853.5	0	1,853.5 to 1,853.5
Thompson	1,909.6	51.3	1,891.8 to 1,927.9

Page Hinkley calibration

Fixed deploy, IID failure labels, 1,150 observations, seeds 1000-1029. The selected threshold is 400; selection criterion is no more than 10% of streams with any alarm in each of two null cells. This finite-sample screen is not a theoretical error bound.

Threshold	Failure rate	Mean alarms	Fraction with any alarm
20	0.1	14.9	1.000
20	0.35	26.5	1.000
50	0.1	3.5	0.967
50	0.35	8.5	1.000
100	0.1	0.7	0.533
100	0.35	2.2	0.867
200	0.1	0	0.000
200	0.35	0.4	0.367
400	0.1	0	0.000
400	0.35	0	0.033
800	0.1	0	0.000
800	0.35	0	0.000

Drift evaluation

Mean cumulative expected pseudo-regret uses all 500 decisions. Realized costs exclude a common censored-delay mask. Terminal observations are included in cost evaluation but not in model learning. The moment-rate control accounts for the synthetic canary multiplier; CI replay controls assume an action-independent label.

Policy	Stationary	Abrupt	Gradual
Static rules	77.1	88.3	94.7
LinUCB	53.5	65.1	68.1
Thompson	66.9	70.8	79.5
PH threshold 50	55	76.8	71.2
PH threshold 400	53.5	65.1	68.1
Scalar moment rate	23.2	163.1	73.8
Rate window 50	34.1	47.5	36.4
Rate window 100	31.5	64.6	34.7

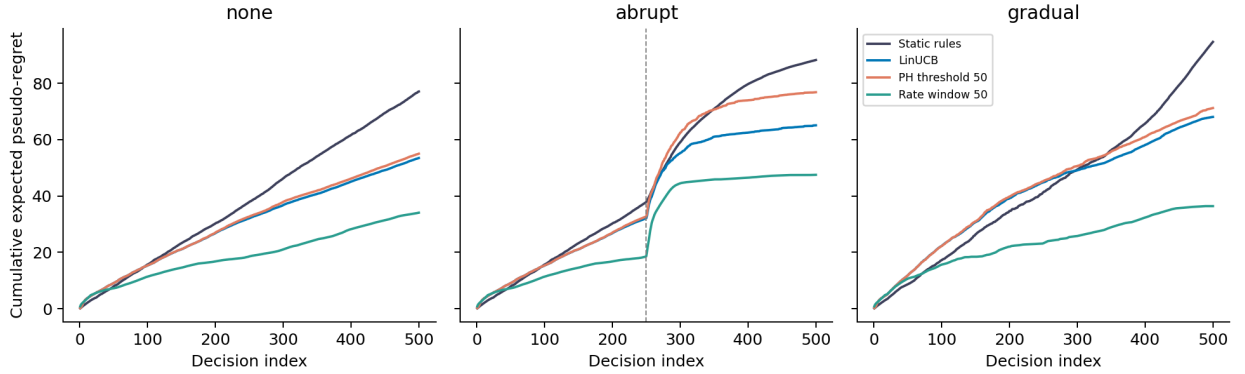


Figure 1: Expected pseudo-regret by decision index

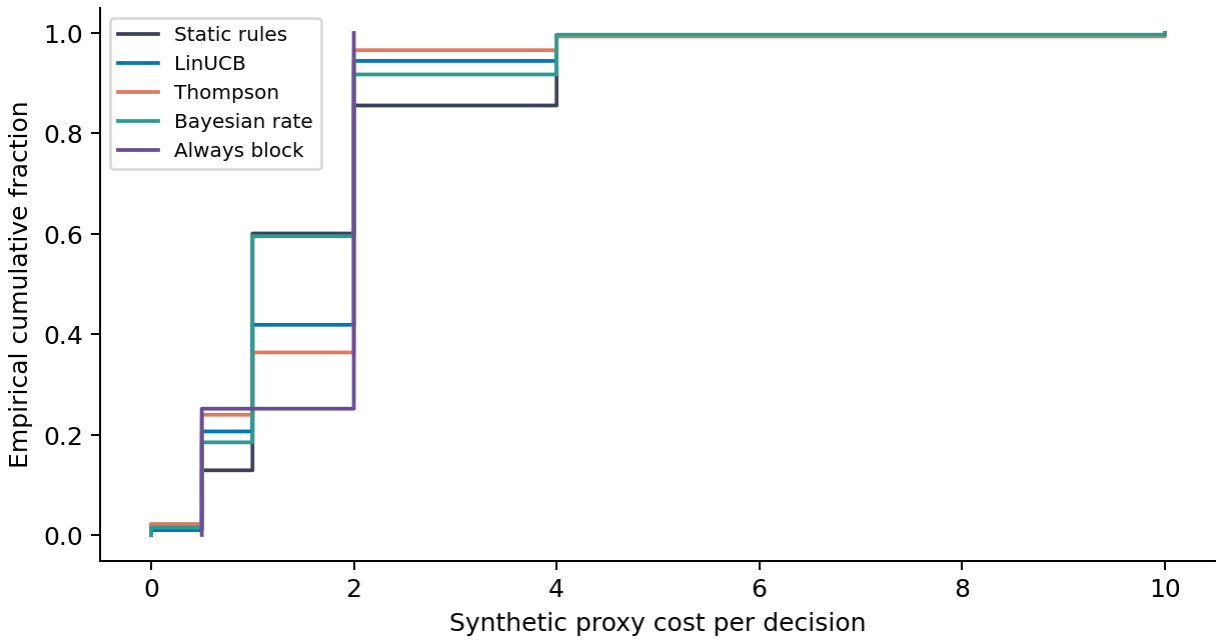


Figure 2: Empirical synthetic cost distribution

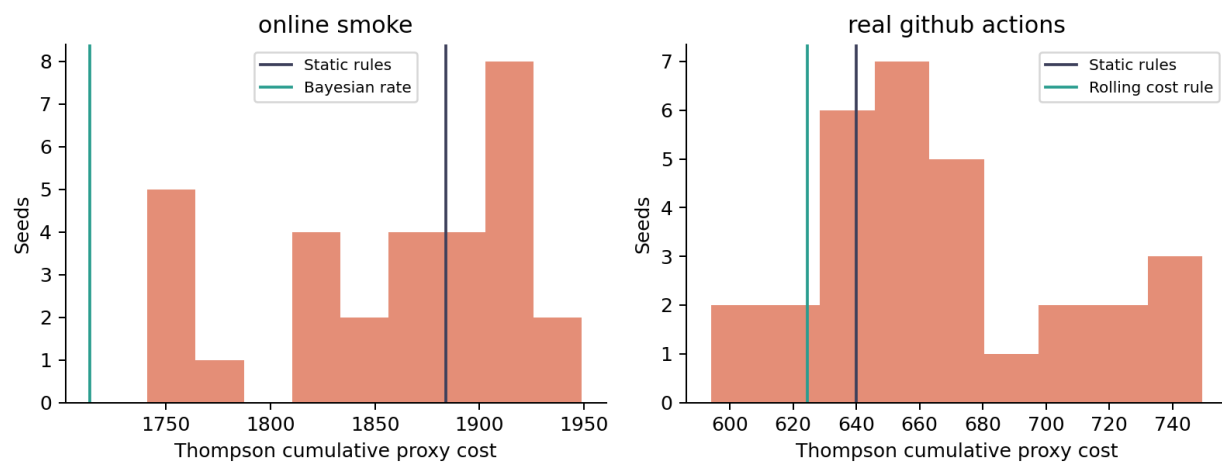


Figure 3: Thompson seed costs and simple comparators

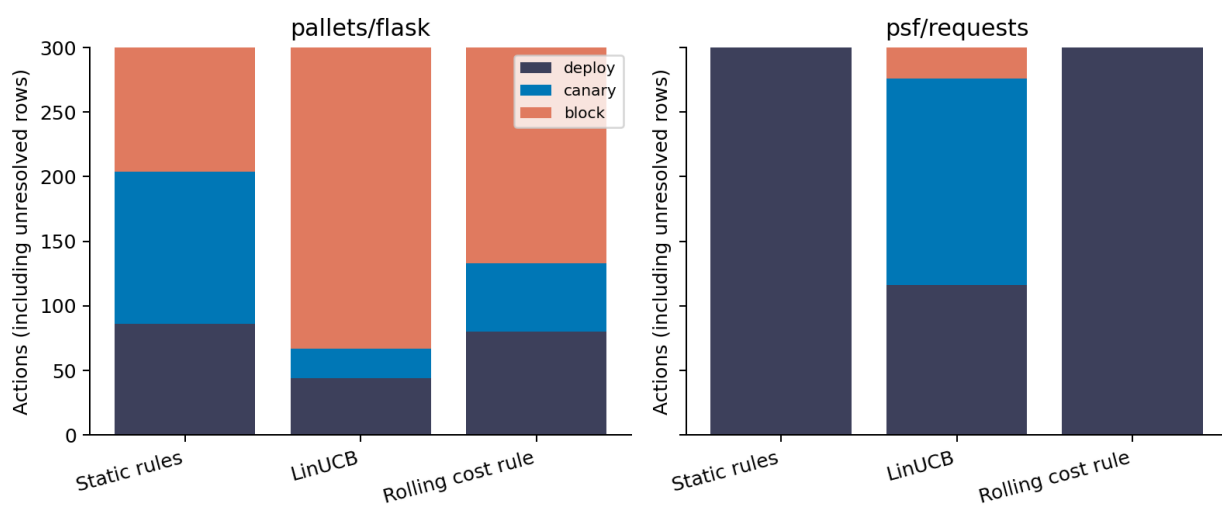


Figure 4: Project-local real-data action counts

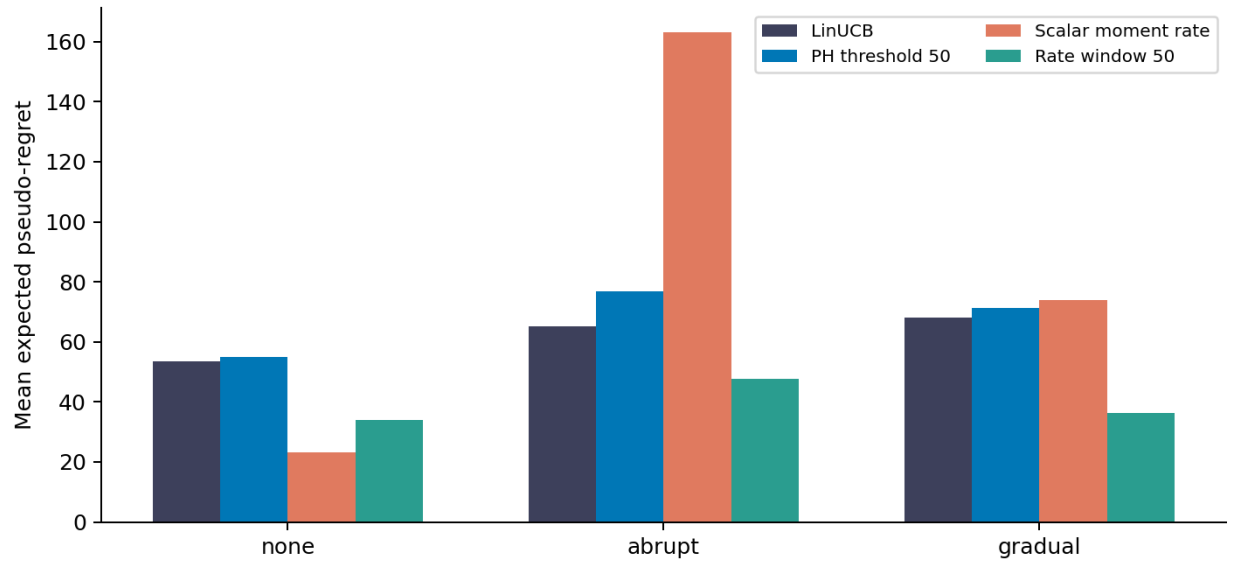


Figure 5: Expected pseudo-regret across drift modes

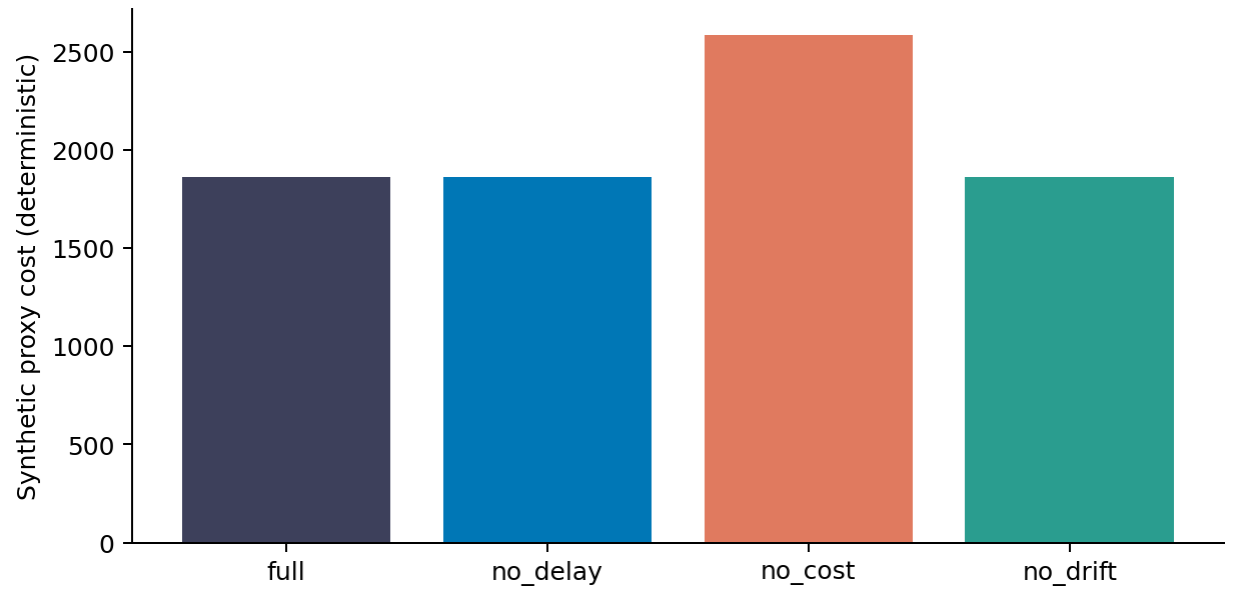


Figure 6: Corrected deterministic ablation costs

Reproducibility

Run the commands in the root README from a clean checkout. The frozen CSV hashes and code hashes are in `corrected/manifest.json`. This run used Python 3.13 and the exact dependency versions in `requirements-audit.txt`. The main build scripts use Pandoc and a TeX installation.

Remaining limitations

The synthetic fixture generator is unavailable. The real export lacks collection provenance and workflow identities; it cannot be interpreted as independent deployment decisions. Deterministic policy repetition does not supply dataset uncertainty. Hyperparameter sensitivity is exploratory on the same export. Real opportunity costs, deployment incidents, action-dependent censoring and policy effects on future releases are unmeasured. The original empirical operating-boundary claim is withdrawn.